SUPPLY CHAIN ANALYTICS PORTFOLIO

Pharmaceutical Inventory Health Monitor

Raw Material to Finished Goods Inventory Risk Analysis
原料至成品三层库存健康度分析

Scope: 80 SKUs | 58 Raw Lots | 30 WIP Batches | 120 Finished Batches

Focus: Inventory valuation, expiry risk, batch traceability, cost efficiency

Tools: Excel (dynamic arrays) · Python (data validation & charting)

Self-directed case study · Simulated dataset

Executive Summary

执行摘要

This case study builds an end-to-end inventory health monitor for a pharmaceutical / nutraceutical manufacturer, covering the full material flow from raw material lots, through work-in-process (WIP), to finished goods shipped to export customers. The objective is to demonstrate how structured inventory analytics turns scattered batch records into a single view of **value at risk, expiry exposure, and batch traceability**.

NZD 15.0M	NZD 200K	NZD 524K
Total Inventory Value 总库存价值	Already Expired 已过期价值	Expiring ≤ 60 days 60天内临期

Key Findings 主要发现

1. NZD 200K of inventory has already expired and requires immediate write-off

17 batches are past their expiry date, tying up capital and creating GMP recall-audit obligations. 17 个批次已过期,需立即处置。

2. A further NZD 524K expires within 60 days – a clear action window

Near-expiry stock flagged across three colour-coded tiers enables prioritised clearance before value is lost. 临期库存按三色分级,可在损失前优先清理。

3. Full batch traceability is achievable from finished goods back to supplier lot

Every finished batch links to its WIP batch and originating raw material lot – the backbone of GMP / Medsafe recall readiness. 每个成品可反查至原料 Lot 与供应商,支撑 GMP 召回。

4. Inventory concentrates in four export customers across CN / AU / US / SEA markets

Customer-level visibility supports demand planning and allocation decisions. 库存集中于四个出口客户,支撑需求计划。

Method & Data Structure

方法与数据结构

The analysis follows an **EDA-first discipline**: the raw dataset was profiled for structure, completeness and business invariants before any calculation or visual was built. All key figures were re-validated in Python directly against the source workbook, rather than trusting on-screen totals.

Three-Tier Inventory Model 三层库存模型

Pharmaceutical inventory cannot be treated as a single flat stock list. Material moves through three distinct states, each tracked at batch / lot level with its own expiry clock:

TIER 1 · RAW MATERIALS (原料)

58 supplier lots

Tracked by Lot number, supplier, category, intake & expiry date, remaining quantity, unit cost. e.g. Vitamin D3, fish oil, krill oil, probiotics.

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TIER 2 · WORK-IN-PROCESS (在制品)

30 WIP batches

Links consumed raw lots to a target finished SKU; tracks charge date, expected completion, and current production stage (e.g. QA testing).

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TIER 3 · FINISHED GOODS (成品)

120 finished batches

Tracked by production batch, source WIP batch, manufacture & expiry date, current quantity, and inventory value to the customer.

Data Quality Principles 数据质量原则

- **Structural blanks preserved as facts** – missing values that carry meaning (e.g. "no batch alert") are kept, not zero-filled.
- **Independent validation** – 80 SKUs, 58 lots, 30 WIP, 120 finished batches confirmed against source.
- **Lot-level expiry tracking** – each batch carries its own expiry clock; aggregate totals never hide an at-risk batch.

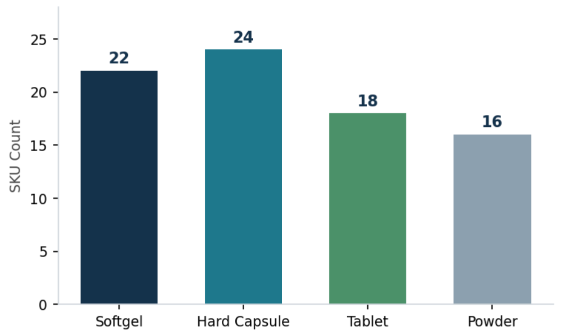
Inventory Health Overview

库存健康总览

80	58	30	120
Total SKUs SKU 总数	Raw Lots 原料 Lot	WIP Batches WIP 批次	Finished Batches 成品批号

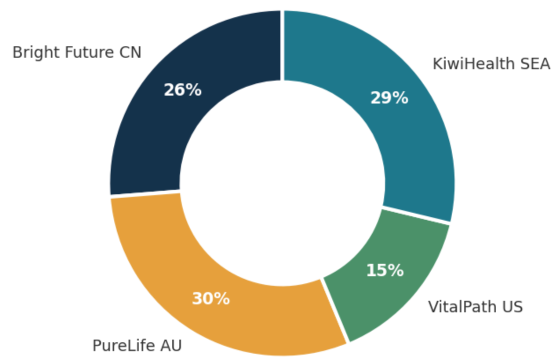
The product portfolio spans four dosage forms and four export customers, giving a balanced but monitorable spread of inventory across markets.

By Dosage Form 按剂型



SKU count across softgel, hard capsule, tablet, powder

By Customer 按客户



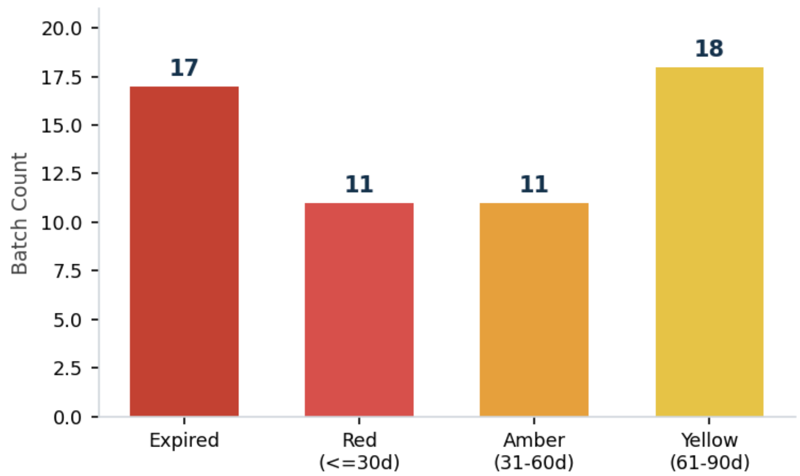
SKU distribution across four export markets

Hard capsules (24 SKUs) and softgels (22) form the bulk of the range, consistent with a nutraceutical contract-manufacturing profile. On the demand side, no single customer dominates: PureLife AU and KiwiHealth SEA are the largest, but all four markets carry meaningful volume – reducing concentration risk but increasing coordination complexity.

Expiry & Risk Exposure

临期与过期风险

A three-tier colour-coded alert system classifies every batch by days-to-expiry, converting raw dates into an actionable risk queue. Across the portfolio, **57 batches carry an expiry flag, representing roughly NZD 1.61M of inventory value** at some stage of risk.



Batch count by alert tier — 17 already expired

Value at Risk 风险价值

NZD 200K	NZD 350K	NZD 174K
Expired (write-off) 已过期	≤ 30 days 30天临期	31–60 days 60天临期

Top High-Risk Batches 高风险批次(节选)

Batch No.	Type	Expiry	Days	Value (NZD)	Action
FG-HG-240926-090	Finished	2026-03-20	-64	10,103	Write-off + recall audit
FG-PW-241003-086	Finished	2026-03-27	-57	9,500	Write-off + recall audit
FG-PW-241007-040	Finished	2026-03-31	-53	7,600	Write-off + recall audit
FG-TB-230419-012	Finished	2026-04-03	-50	5,412	Write-off + recall audit
LOT-P008-2505-037	Raw	2026-05-02	-21	21,750	Write-off + recall audit
LOT-P002-2405-008	Raw	2026-05-05	-18	12,000	Write-off + recall audit

Why this matters for a pharma supply chain

Expired stock is not just lost value — in a GMP-regulated environment each expired batch may trigger quarantine, documented disposal and a recall-audit trail. Surfacing these batches early, ranked by days-to-expiry and value, lets the team act inside the window rather than discovering losses at stock-count.

Batch Traceability & Recommendations

批次可追溯性与建议

Given any finished batch number, the model reconstructs the full production chain back to the supplier lot – the core capability behind GMP / Medsafe recall readiness.

Worked Example 追溯链示例

FINISHED GOODS 成品

FG-SG-260522-031

Vitamin D3 5000IU Softgel · Qty 2,997 · Mfg 2026-05-22

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← FROM WIP BATCH 来自 WIP

WIP-2605-002

Stage: QA testing · Qty 13,030 · 2026-05-10

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← FROM RAW LOT 来自原料 LOT

LOT-P009-2604-041

DSM Nutritional Products / Vitamin D3 · Qty 300 · 2026-04-24

Recommendations 建议

- **Immediate:** Write off the 17 expired batches (NZD 200K) with documented disposal, and quarantine the two expired raw lots pending supplier review.
- **Short-term:** Action the NZD 350K of stock expiring within 30 days – prioritise by value and customer commitment.
- **Process:** Embed the three-tier expiry alert into a weekly review so near-expiry stock is caught inside the action window, not at quarterly count.
- **System:** Carry the lot-to-finished traceability link into the ERP / MRP rollout so recall readiness is built in, not bolted on.

Data & Scope Disclaimer 数据与范围声明

This is an independent, self-directed case study built on a **simulated dataset** created for skills demonstration. It does not use any real company's data and does not represent actual professional or client work.

本作品为独立自主案例研究,使用**自制模拟数据集**,仅用于能力展示,不涉及任何真实公司数据,亦不代表真实职业或客户工作经历。